



Generating solutions to far analogies makes individuals tend to categorize information based on thematic relations

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Abstract

This study aimed to explore whether solving distant analogies makes individuals tend to categorize information based on either taxonomic or thematic relations. In the study, one group of participants solved far analogies (far analogy group), while another group solved near analogies (near analogy group). Then, all participants completed the triad task which is the task of measuring the propensity to classify. The research findings revealed that, regardless of whether the object of classification was the artifact or natural object, the far analogy group exhibited a higher percentage of thematic responses than the near analogy and control group in the triad task. The present study demonstrated that solving far analogies could make individuals tend to categorize information based on thematic relations.

Keywords Analogy · Far analogy · Thematic relation · Taxonomic relation · Semantic categorization

Introduction

Organizing our knowledge of the world into useful categories is one of the brain's most fundamental processes. Categories are sets or classes of items or entities that are similar or connected in some meaningful manner (Murphy 2002). Two types of relation, taxonomic and thematic, have been widely proposed to provide structure to our stored, semantic knowledge about categories of concrete objects (Estes et al. 2011; Lawson et al. 2017; Mirman et al. 2017). Thematic relations have been defined as any temporal, spatial, causal, or functional relation between things (Lin and Murphy 2001). More specifically, things are thematically related if they perform complementary roles in the same scenario or event (Estes et al. 2011; Lin and Murphy 2001). For example, the objects “cow” and “milk” are related by a production theme, cows are producers and their milk is the product, and they perform complementary thematic roles. Taxonomic

relations have been defined simply by using established categories in the knowledge system (Collins and Quillian 1969). To be specific, things are taxonomically related if they have common properties or perceptual features (Estes et al. 2011). For example, “whales” and “horses” share important features (e.g., being warm-blooded and bearing live offspring) and hence belong to the “mammals” category; “pizza” and “chips”, due to their shared property of being edible, are both members of the “food” category.

Studies have demonstrated that categorizing information based on thematic relations or taxonomic relations has different effects on cognitive processing in individuals. Compared to classification based on taxonomic relations, classification based on thematic relations has been found to increase subjectively perceived similarity (Golonka and Estes 2009; Simmons and Estes 2008; Wisniewski et al. 1999), facilitate the learning and memory of texts, and stories (Seifert et al. 1986), aid memory (Estes, 2003), influence language comprehension (Estes and Jones 2006), and support reasoning (Chaigneau et al. 2009; Saalbach and Imai 2007).

The tendency of individuals to categorize information based on thematic or taxonomic relations is influenced by various factors. Research has shown that compared with initiating local processing, initiating global processing will lead individuals to classify information based on thematic relations (Guest et al. 2016). Guest et al. (2016) evoked global

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or local processing in the two groups and then completed the triad task. This task presented a set of triads in which there is a base item (e.g. dog) and two comparison items. Participants needed to choose whether the taxonomically related item (e.g. cat) or the thematically related item (e.g. bone) should belong to the same category as the base item. The results found that compared with the local processing group, the global processing group has a higher proportion of selecting the thematic relation option. Additionally, research has shown that cultural type, thought type, age, and abstractness of material, also influence whether individuals categorize information based on thematic or taxonomic relations. Unconscious thoughts tend to categorize information based on thematic relations, whereas conscious thoughts tend to categorize information based on taxonomic relations (Li et al. 2022). East Asians tend to categorize information based on thematic relations, whereas Europeans and Americans tend to categorize information based on taxonomic relations (Nisbett and Masuda 2003). Adults tend to categorize information based on thematic relations, whereas children tend to categorize based on taxonomic relations (Berger and Donnadieu 2006). Categorizing concrete concepts relies more strongly on taxonomic relations, whereas categorizing abstract concepts relies more powerfully on thematic relations (Crutch and Warrington 2010). In addition to the above factors, whether solving analogies also affects the classification tendency of individuals is worthy of further exploration.

An analogy can be defined as a mapping from a source domain to a target domain (Gentner 1983; Gick and Holyoak 1980; Holyoak 2012). The amount of overlap between domains has been used to characterize analogies along a continuum from “near” to “distant” (Gentner 1983; Ward 1998). As to details, the source and target domains can be perceptually and/or conceptually similar in a near analogy, or quite different in a far analogy. That is, if you are walking near the harbor and you observe that catching salmon with a drift net is similar to catching shrimp with a trawl net, you have made a near analogy. If, in walking from the harbor back to your dorm, you pass the Math building and observe that using nets to catch fish is similar to using prediction intervals to predict future sample characteristics, you have made a far analogy. Research has shown that solving distant analogies is more beneficial in discovering structural relational similarities between things, as compared to solving near analogies (Vendetti et al. 2014). Vendetti et al. (2014) conducted an experiment wherein the participants were divided into two groups. One group was given a series of near analogy problems, while the other group was given a series of far analogy problems. Near analogies are those that

can be solved by matching the surface relationship similarity, such as “the furnace burns coal as a woodstove burns wood.” Far analogies, on the other hand, require the evaluation or generation of a more abstract relation that bridges the domains, such as “the furnace burns coal as the stomach ‘burns’ food.” After solving all the analogy problems, the participants were asked to complete the picture-mapping task. The study found that participants solving distant analogies gave more structural relational responses. In addition, various studies have also shown that compared to solving near analogies, solving distant analogies is closely associated with a reduction in belief-based responses in transitive inference (Andrews et al. 2019), creative processing (Green 2016; Srinivasan et al. 2018) and analogical retrieval (Goldwater and Jamrozik 2019), among others. A question arises then, as to whether solving distant analogies makes individuals tend to categorize information based on either taxonomic or thematic relations.

Unlike categorizing information based on taxonomic relations, categorizing information based on thematic relations is closely related to abstract processing (Estes and Jones 2006; Mirman et al. 2017). Several studies have shown that solving far analogies is beneficial for abstract understanding (e.g., Chaxel 2015; Green et al. 2012). For instance, Chaxel (2015) found that solving distant analogies allows individuals to create high-level explanatory statements about people’s behavior. Additionally, categorizing information based on thematic relations involves identifying overall relationships among things, rather than discovering common detailed features required by taxonomic relations (Mirman et al. 2017). Research has also indicated that solving far analogies facilitates relational integration (Andrews and Bohadana 2018; Du and Sun 2022). For example, Andrews et al. (2018) found that solving far analogies is beneficial for the integration of complex relationships between letters. Therefore, the present study hypothesized that solving far analogies makes individuals tend to categorize information based on thematic relations.

The current study involved manipulating near and far analogies through a verbal analogy task (Vendetti et al. 2014; Andrews et al. 2019). Participants solved four-term verbal analogy problems, with half of the participants receiving near analogies and the other half receiving far ones. They then completed a task to measure classification tendency. If solving far analogies could make individuals tend to categorize information based on thematic relations, then the percentage of thematic responses within the distant analogy group in the triad task should be higher, regardless

of whether the object of classification is the artifact (Experiment 1) or natural object (Experiment 2).

Experiment 1

In Experiment 1, preference for thematic or taxonomic relations was measured using the triad task (e.g., Guest et al. 2016; Lin et al. 2001; Simmons et al. 2008). If solving far analogies makes individuals tend to categorize information based on thematic relations, the percentage of thematic responses which pertain to the distant analogy group in the triad task should be higher.

Methods

Participants and design

We used G*power to conduct the minimum sample size (Faul et al. 2007). The effect size of this experiment was set at $d=0.5$, and the power was set at 0.80 ($\alpha=0.05$). The results revealed that the minimum sample size for the experiment was 128. Hence, in total, 131 university students (62 male and 69 female, mean age = 20.36 years, $SD=0.57$) participated in Experiment 1. They were randomly divided into two groups: a far analogy group (31 male and 35 female) and a near analogy group (31 male and 34 female). All participants signed an informed consent form and were paid accordingly at the end of the experiment. In addition, before the formal experiment, all participants completed an abridged version of Raven's Advanced Progressive Matrices (RAPM; Arthur et al. 1999). The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha=0.05$, both the data from the far-analogy group ($p=0.17$) and the near-analogy group ($p=0.09$) conform to a normal distribution. The results of independent sample t-tests showed that the score of participants in the far analogy group ($M=7.12$, $SD=2.12$) on the RAPM test was not significantly different from those in the near analogy group ($M=6.74$, $SD=2.10$), $t(129)=1.01$, $p=0.31$.

This experiment adopted a single-factor between-participants design. The type of analogy question (far analogy, near analogy) was set as the between-participants variables. The measurement index of the dependent variable was the percentage of thematic responses in the triad task.

Materials and procedure

The material for the analogical task was acquired from verbal analogy problem sets assembled by Ichien and colleagues (2020). To ensure the cross-cultural validity of the

experiment, we adopted the research methodology suggested by Ember et al. (2001) and undertook the translation and back-translation process of the word material. Two seasoned translators with 6 and 8 years of experience were employed for translating the materials independently. The discrepancies between the two translations were carefully examined and reconciled, culminating in the first draft in Chinese. Subsequently, an expert in psychology with a background in English translated the first draft from Chinese back into English. Finally, we compared the back-translated version with the original version and adjusted the Chinese first draft to create the Chinese version employed in the experiment. In a similar vein, the word material for the triad task was obtained from Simmons et al. and Lin et al. (Lin and Murphy 2001; Simmons et al. 2008). Once again, we followed the same translation and back-translation process discussed earlier and generated the Chinese version for this task.

In preparation for the formal experiment, participants were given prior information that they would be required to complete an analogy task. In the formal experiment, participants were randomly assigned to one of two groups, with one group completing 40 far analogy problems (e.g., stove: coal:: gut: food), and the other group completing 40 near analogy problems (e.g., stove: coal:: wood stove: charcoal). The order of presentation for the analogy problems was decided through computer-generated randomization. Before each analogy problem display, a gaze point of 500 ms duration was presented to the participant. The presented stimulus comprised of “A: B:: C: ?” (where A: B always appeared above C: ?). At this point, participants were instructed to think about the appropriate answer to be filled in the “?” position. Once they concluded, they wrote the answer on the response sheet provided and proceeded by pressing the spacebar for the next question.

After solving all analogy problems, we asked participants to complete the triad task. In this task, a series of items was presented in sequence on a computer screen, with the presentation order randomly generated by the computer. Each item consists of a benchmark word (such as “lamp”)

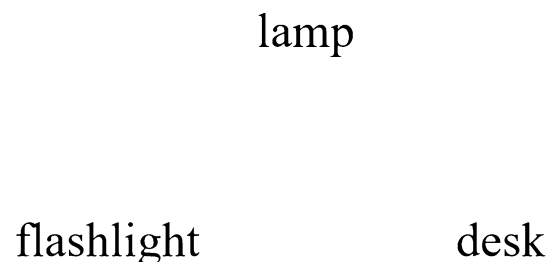


Fig. 1 The example of the triad items

and two comparative words, one of which is taxonomically related to the base word (such as “flashlight”, both it and the lamp have the attribute characteristics of luminous lighting) and the other thematically related to the base word (such as “desk”, it and the lamp play a complementary role in the theme scene of “learning” or “handle official business”) (see Fig. 1). Participants were instructed to determine which comparative word should be grouped with the benchmark word. The proportion of choosing thematically related comparative words made by the participants served as the measure of the participants’ classification tendency. Half of the thematically related word is presented on the left side of the item and the other half on the right side.

Results and discussion

We used JASP software (Wagenmakers et al. 2018) to perform the Independent sample t-test and Bayesian analysis (default prior, Jamil et al. 2017) on the data respectively. The Bayes factor (BF) can quantify the strength of the evidence supporting the alternative or null hypothesis (Kruschke 2013). Usually, BF_{10} indicates the strength of support for the alternative hypothesis, and BF_{01} indicates the strength of support for the null hypothesis. When the value of BF_{10} is greater than 1, the alternative hypothesis is supported (significant interaction or difference between groups), and the larger the value the higher the strength of support. Similarly, when the value of BF_{01} is greater than 1, the null hypothesis is supported (the interaction or the difference between groups is not significant), and the larger the value the higher the strength of support (Rouder et al. 2009). The specific standard: $1 < BF \leq 3$ weak, $3 < BF \leq 20$ positive, and $20 < BF \leq 150$ strong supports the alternative or null hypothesis (Wagenmakers 2007).

We calculated the accuracy of the near and far analogy groups in solving analogy problems. The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha = 0.05$, both the data from the far-analogy group ($p = 0.20$) and the near-analogy group ($p = 0.20$) conform to a normal distribution. Subsequently, the independent sample t-tests were conducted with the type of analogy problem with grouping conditions (far analogy, near analogy). The results showed that the correct rate of the near analogy group ($M = 0.70$, $SD = 0.16$) was significantly higher than that of the far analogy group ($M = 0.51$, $SD = 0.17$), $t(129) = 6.54$, $p < 0.001$, Cohen’s $d = 1.14$, $BF_{10} = 6.83 \times 10^6$ in favor of an effect. These results replicated the findings of Green et al. (2012) and Vendetti et al. (2014), which suggested that the manipulation of analogy problem types in this study is effective.

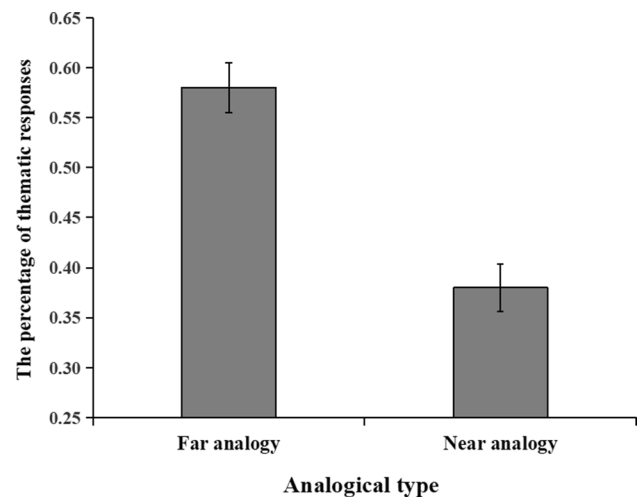


Fig. 2 The percentage of thematic responses of the near and far analogy group (Error bars represent the average standard errors)

We calculated the percentage of thematic responses of the near and far analogy group in the triad task (see Fig. 2). The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha = 0.05$, both the data from the far-analogy group ($p = 0.09$) and the near-analogy group ($p = 0.20$) conform to a normal distribution. Subsequently, the independent sample t-tests were conducted with the analogy problem type with grouping conditions (far analogy, near analogy). The results showed that the percentage of thematic responses in the triad task of the far analogy group ($M = 0.58$, $SD = 0.21$) was significantly higher than that of the near analogy group ($M = 0.38$, $SD = 0.19$), $t(129) = 5.91$, $p < 0.001$, Cohen’s $d = 1.03$, $BF_{10} = 3.85 \times 10^5$ in favor of the effect. The results revealed that solving far analogies could make individuals tend to categorize information based on thematic relations.

Experiment 2

In Experiment 1, we employed a real-world version of the triad task, in which all the objects were artifacts (human-made items such as cars). In Experiment 2, we added natural objects (naturally occurring objects or “natural kinds” such as cats) to the classification task and further examined whether solving far analogies still could make individuals tend to categorize information based on thematic relations when the object of classification was a natural object, based on a replication of the findings of Experiment 1. Additionally, in Experiment 2, we added a group of participants who

completed a word rating task as a control group. If solving far analogies could make individuals tend to categorize information based on thematic relations, regardless of whether the objects being classified are artificial or natural, compared to participants in the analogy group and control group, the percentage of thematic responses which pertain to the far analogy group in the triad task should be higher.

Methods

Participants and design

We used G*power to conduct the minimum sample size (Faul et al. 2007). The effect size of this experiment was set at $f=0.25$, and the power was set at 0.80 ($\alpha=0.05$). The results revealed that the minimum sample size for the experiment was 159. Hence, in total, 199 university students (100 males and 99 females, mean age = 21.15 years, $SD=0.57$) participated in Experiment 2. They were randomly divided into three groups: a far analogy group (32 males and 34 females), a near analogy group (31 males and 36 females), and a control group (36 males and 30 females). All participants signed an informed consent form and were paid accordingly at the end of the experiment. In addition, before the formal experiment, all participants completed an abridged version of Raven's Advanced Progressive Matrices (RAPM; Arthur et al. 1999). The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha=0.05$, both the data from the far-analogy group ($p=0.09$), the near-analogy group ($p=0.09$), and the control group ($p=0.20$) conform to a normal distribution. The results of one-way ANOVA showed that the score of participants in the far analogy group ($M=7.02$, $SD=2.17$) on the RAPM test was not significantly different from those in the near analogy group ($M=7.33$, $SD=2.30$), and the control group ($M=6.79$, $SD=2.14$), $F(2, 196)=1.01$, $p=0.37$.

This experiment adopted a single-factor between-participants design. The type of analogy question (far analogy, near analogy, control) was set as the between-participants variables. The measurement index of the dependent variable was the percentage of thematic responses in the triad task.

Materials and procedure

Experiment 2 used the triad task in which half of the items are replaced with natural objects. The replacement word material was still obtained from Simmons et al. (2008) and Lin et al. (2001). Similarly, we invited the experts above to translate and back-translate the word material in turn, resulting in the Chinese version used in the experiment.

A control was added to Experiment 2, and the other experimental procedures were similar to that of Experiment 1. The formal experiment began by asking one group of participants to solve 40 far analogy problems and another group of participants to solve 40 near analogy problems. Participants in the control condition rated the pleasantness of sixty trait adjectives sourced from Anderson (1968). Half of them are positive words (e.g. generous) and the other half are negative words (e.g. boastful). The presentation order of words was randomly generated by the computer. Participants were requested to rate each adjective in terms of its pleasantness on a scale from 1 to 7, where 1 represented very unpleasant and 7 represented very pleasant.

After completing the analogy task (for the distant and near analogy groups) or word scoring task (for the control group), participants were asked to complete a balanced version of the triad task. Under the three experimental conditions, each group had half of the participants complete the triad items consisting of artifacts first, followed by the triad items consisting of natural objects. The other half of the participants completed the triad items consisting of natural objects first and then the triad items consisting of artifacts. The triad items were presented in the same way as in Experiment 1.

Results and discussion

As with Experiment 1, we used JASP software to perform the Independent sample t-test, One-way ANOVA, and Bayesian analysis of the data respectively.

We calculated the accuracy of the near and far analogy groups in solving analogy problems. The results of the Kolmogorov–Smirnov test indicated that, at a significance level

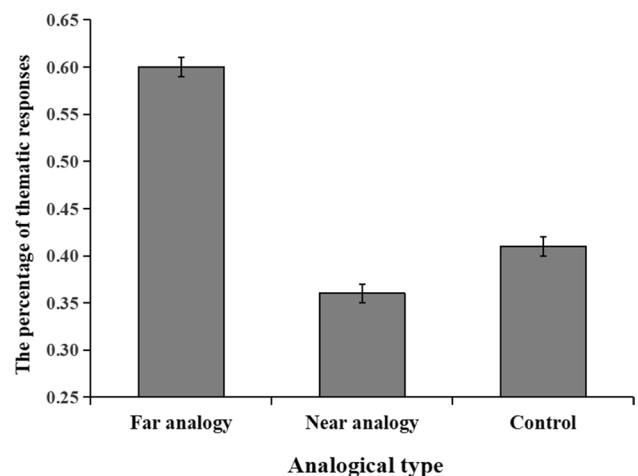


Fig. 3 The percentage of thematic responses of the control, near, and far analogy groups (Error bars represent the average standard errors)

of $\alpha = 0.05$, both the data from the far-analogy group ($p = 0.20$) and the near-analogy group ($p = 0.20$) conform to a normal distribution. Following this, the independent sample t-tests were conducted with the type of analogy problem with grouping conditions (far analogy, near analogy). The results showed that the correct rate of the near analogy group ($M = 0.69$, $SD = 0.17$) was significantly higher than that of the far analogy group ($M = 0.52$, $SD = 0.15$), $t(131) = 5.93$, $p < 0.001$, Cohen's $d = 1.03$. These results replicated the findings of Green et al. (2012) and Vendetti et al. (2014), which suggested that the manipulation of analogy problem types in this study is effective.

We calculated the percentage of thematic responses of the control, near analogy, and far analogy groups in the triad task (see Fig. 3). The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha = 0.05$, both the data from the far-analogy group ($p = 0.20$), the near-analogy group ($p = 0.20$), and the control group ($p = 0.20$) conform to a normal distribution. Then, the analogy problem type as the independent variable to perform an analysis of variance (ANOVA). The results showed that the main effect of analogical problem types is significant, $F(2, 196) = 167.60$, $p < 0.001$, $\eta^2 = .63$, $BF_{10} = 4.61 \times 10^{39}$ in favor of the effect. Post-hoc tests with Bonferroni corrections for multiple comparisons showed that the percentage of thematic responses in the triad task of the far analogy group ($M = 0.60$, $SD = 0.19$) was significantly higher than that of the near analogy group ($M = 0.36$, $SD = 0.20$), $p < 0.001$, $BF_{10} = 3.02 \times 10^{32}$ in favor of the effect; and the control group ($M = 0.41$, $SD = 0.20$), $p < 0.001$, $BF_{10} = 1.59 \times 10^{22}$ in favor of the effect. In addition, there was a significantly different between the results of the control and near analogy groups, $p < 0.001$, $BF_{10} = 180.85$ in favor of the effect.

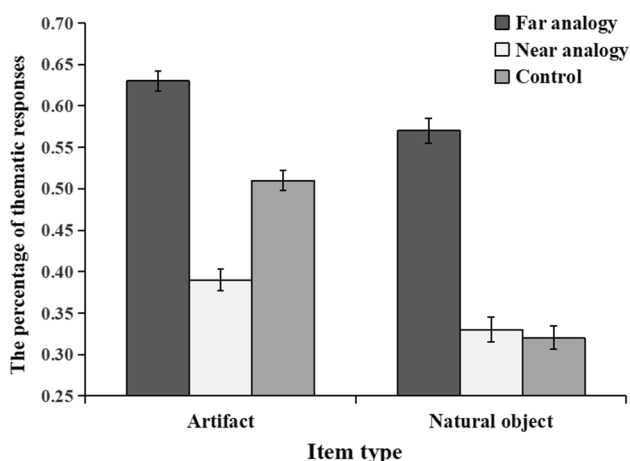


Fig. 4 The percentage of thematic responses of the control, near, and far analogy groups in the triad items which are artifacts or natural objects (Error bars represent the average standard errors)

More importantly, we calculated the percentage of thematic responses of the control, near analogy, and far analogy groups in the triad items that solely consist of natural objects (see Fig. 4). The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha = 0.05$, both the data from the far-analogy group ($p = 0.20$), the near-analogy group ($p = 0.17$), and the control group ($p = 0.09$) conform to a normal distribution. Then, the analogy problem type as the independent variable to perform an analysis of variance (ANOVA), the main effect of the analogy problem type was significant, $F(2, 196) = 68.27$, $p < 0.001$, $\eta^2 = 0.41$, $BF_{10} = 1.18 \times 10^{20}$ in favor of the effect. Post-hoc tests with Bonferroni corrections for multiple comparisons showed that the percentage of thematic responses of the far analogy group ($M = 0.57$, $SD = 0.15$) was also significantly higher than that of the near analogy group ($M = 0.33$, $SD = 0.13$), $p < 0.001$, $BF_{10} = 3.57 \times 10^{14}$ in favor of the effect; and the control group ($M = 0.32$, $SD = 0.14$), $p < 0.001$, $BF_{10} = 4.21 \times 10^{14}$ in favor of the effect. Similarly, we calculated the percentage of thematic responses of the control, near analogy, and far analogy groups in the triad items that solely consist of artifacts (see Fig. 4). The results of the Kolmogorov–Smirnov test indicated that, at a significance level of $\alpha = 0.05$, both the data from the far-analogy group ($p = 0.20$), the near-analogy group ($p = 0.20$), and the control group ($p = 0.07$) conform to a normal distribution. Subsequently, the analogy problem type as the independent variable to perform an analysis of variance (ANOVA), the main effect of the analogy problem type was significant, $F(2, 196) = 116.37$, $p < 0.001$, $\eta^2 = 0.54$, $BF_{10} = 7.18 \times 10^{30}$ in favor of the effect. Post-hoc tests with Bonferroni corrections for multiple comparisons showed that the percentage of thematic responses of the far analogy group ($M = 0.63$, $SD = 0.09$) was significantly higher than that of the near analogy group ($M = 0.39$, $SD = 0.08$), $p < 0.001$, $BF_{10} = 6.90 \times 10^{27}$ in favor of the effect; and the control group ($M = 0.51$, $SD = 0.09$), $p < 0.001$, $BF_{10} = 2.23 \times 10^8$ in favor of the effect. The results indicated that, whether classifying natural or artificial objects, solving far analogies could make individuals tend to categorize information based on thematic relations.

General discussion

By manipulating the analogy problem type that participants solved, this study explored whether solving far analogies makes individuals tend to categorize information based on thematic relations. The results found that regardless of whether the object of classification was the artifact or natural object, the far analogy group exhibited a higher percentage of thematic responses in the triad task. Consequently, these

findings revealed that solving far analogies could make individuals tend to categorize information based on thematic relations.

This study expanded the research on when individuals classify information based on thematic relations. Previous studies have shown that global processing, unconscious thinking, adulthood, East Asian culture, and abstract concepts are all closely associated with individuals categorizing information based on thematic relations (e.g., Berger et al. 2006; Crutch et al. 2010; Guest et al. 2016; Li et al. 2022; Nisbett et al. 2003). The present study further found that solving far analogies could make individuals tend to classify information according to thematic relations.

This study extended the research on the effect of solving far analogies on individual cognitive processing. Research has shown that solving far analogies is intimately associated with reducing belief-based responses in transitive inference, relational integration, creative thinking, abstract understanding, and the discovery of deep structural relationships (e.g., Andrews et al. 2019; Andrews et al. 2018; Chaxel 2015; Green 2016; Vendetti et al. 2014). The present study further found that solving far analogies could make individuals tend to classify information according to thematic relations.

We believe that solving far analogies could make individuals tend to categorize information based on thematic relationships because solving far analogies could activate a relational mindset (e.g., Simms et al. 2019; Vendetti et al. 2014). Categorizing information based on category relationships requires strict adherence to taxonomic hierarchies while categorizing information based on thematic relationships requires attention to the complementary relationships between objects in function or role (Estes et al. 2011). Existing research has shown that solving far analogies activates people's relational mindset (e.g., Simms et al. 2019; Vendetti et al. 2014). When a relational mindset is activated, people tend to seek out and prioritize relational information (e.g., Holyoak and Thagard 1996; Simms et al. 2019). Therefore, solving far analogies makes individuals tend to categorize information based on thematic relationships. Additionally, solving far analogies is beneficial to reducing belief-based responses in transitive inference, relational integration, and the discovery of deep structural relationships, which may also be closely related to the activation of a relational mindset. Of course, further research is needed to test this hypothesis.

It should be clarified in this study that although previous studies have shown that East Asians tend to categorize information based on thematic relationships, the reason why participants in the near analogy group and control group (who are East Asians) in this study tended to categorize information based on taxonomic relations may be due to the different presentation forms of the objects in the two studies.

In the research conducted by Nisbett (2003) and others, the classified objects were presented in the form of drawings, while in this research paradigm, they were presented in the form of words. Studies have shown that connections between words are more difficult to extract from memory than connections between drawings (e.g., Van der Cruyssen et al. 2021). And research has also shown that groups of words with thematic relations are also more difficult to extract from memory compared to groups of words with taxonomic relations (Mirman et al. 2017). Therefore, participants in the near analogy group and control group in this study tended to categorize information based on taxonomic relations. In future research, the presentation of classification objects in the form of drawings should be adopted to further investigate whether solving near analogies could make individuals tend to classify information based on taxonomic relations.

Future studies should select participants from different cultural backgrounds, and examine whether individuals from different cultural backgrounds would tend to categorize information based on thematic relationships when they solved far analogies. Among the studies that have been conducted, Nisbett et al. (2003) found that Europeans and Americans tend to categorize information based on category relations. Moreover, Masuda et al. (2001) also found that Europeans and Americans tend to process information locally. Local processing could make individuals tend to categorize information based on taxonomic relations (Guest et al. 2016). The participants in this study were individuals from East Asian culture (Chinese university students), and future studies should select individuals from Euro–American cultures as participants for further examination.

Future studies should examine whether participants with a typical analytic cognitive style also tend to categorize information based on thematic relationships when they solved far analogies. Individuals with a holistic cognitive style tend to consider objects as a whole and are adept at identifying the global attributes of objects. On the other hand, those with an analytic cognitive style tend to consider objects as sets of parts and are adept at identifying the local attributes of objects (Zhou et al. 2015). The focus of categorizing information based on thematic relationships is to discover the overall relationship between things. Future studies should select participants with a typical analytic cognitive style for further examination.

Appendix 1

See Tables 1 and 2.

Table 1 Verbal analogy problem sets

Analogy source		Far analogy		Near analogy	
Answer	Riddle	Key	Lock	Solution	Problem
Ash	Fireplace	Lint	Pocket	Soot	Chimney
Aspirin	Pain	Muffler	Noise	Antacid	Heartburn
Baker	Cake	Scientist	Discovery	Chef	Meal
Basket	Picnic	Holster	Gun	Lunchbox	Lunch
Basketball	Hoop	Traveler	Destination	Soccer ball	Goal
Blindness	Sight	Poverty	Money	Deafness	Hearing
Blizzard	Snowflake	Army	Soldier	Monsoon	Raindrop
Bracelet	Wrist	Moat	Castle	Ring	Finger
Burger	Bun	Book	Cover	Sub	Roll
Cleanser	Face	Absolution	Sinner	Soap	Body
Eraser	Pencil	Amnesia	Memory	Whiteout	Pen
Father	Son	Inventor	Invention	Mother	Daughter
Sugar	Coffee	Incentive	Deal	Rock candy	Tea
Foresight	Future	X-ray	Bone	Hindsight	Past
Flock	Goose	Constellation	Star	Wolfpack	Wolf
Foundation	House	Premise	Argument	Base	Structure
Furnace	Coal	Stomach	Food	Woodstove	Wood
Hoof	Hoofprint	Introduction	Impression	Paw	Pawprint
Saxophone	Jazz	Brush pen	Calligraphy	Harmonica	Blues
Immunization	Disease	Forewarning	Surprise	Vaccination	Infection
Jacket	Zipper	Wound	Suture	Overcoat	Button
Ketchup	Tomato	Fuel	Petroleum	Guacamole	Avocado
Kitten	Cat	Spark	Fire	Puppy	Dog
Knee	Kneepad	Snail	Shell	Elbow	Elbow pad
Lambchop	Lamb	Chapter	Book	Porkchop	Pig
Landscaper	Lawn	Stylist	Hair	Gardener	Garden
Launchpad	Helicopter	Diving board	Diver	Runway	Airplane
Law school	Lawyer	Vineyard	Wine	Med school	Doctor
Multiplication	Product	Brewing	Beer	Addition	Sum
Nose	Scent	Antenna	Signal	Tongue	Taste
Orchard	Apple	Neighborhood	Apartment	Grove	Orange
Painting	Canvas	Birthmark	Skin	Chinese painting	Rice paper
Pen	Pig	Reservoir	Water	Coop	Chicken
Rectangle	Perimeter	Nation	Border	Circle	Circumference
Revising	Manuscript	Evolving	Species	Editing	Story
Multiplication	Product	Brewing	Beer	Addition	Sum
Thermometer	Temperature	Polygraph	Honesty	Barometer	Pressure
Train	Track	Signal	Wire	Trolley	Rail
Watermelon	Rind	Cigarette	Butt	Orange	Peel

Table 2 The triad items used in Experiments 1 and 2

Experiments 1			Experiments 2		
Base item	Taxonomic item	Thematic item	Base item	Taxonomic item	Thematic item
Lamp	Flashlight	Desk	Lamp	Flashlight	Desk
Tent	Hut	Camp	Tent	Hut	Camp
Ship	Yacht	Sailor	Ship	Yacht	Sailor
Airplane	Car	Airport	Airplane	Car	Airport
Toothbrush	Hairbrush	Denture	Toothbrush	Hairbrush	Teeth
Movie theatre	Opera house	Popcorn	Movie theatre	Opera house	Popcorn
Movie	Documentary	Producer	Movie	Documentary	Producer
Cake	Cookie	Birthday	Dog	Cat	Bone
Shirt	Jacket	Tie	Bee	Butterfly	Honey
Ambulance	Fire truck	Stretcher	Cow	Buffalo	Farm
Diamond ring	Bracelet	Engagement	Penguin	Goose	The antarctic
Police car	Sedan	Police officer	Panda bear	Grizzly bear	Bamboo
Chalk	Marker	Blackboard	Spider	Wasp	Spider web
Ambulance	Fire truck	Stretcher	Squirrel	Rat	Nut

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Data availability The datasets generated for this study are available upon request to the corresponding authors.

Declarations

Conflict of interest The authors declared no potential conflicts of interest concerning the research, authorship, and/or publication of this article.

Human participants The study was conducted after obtaining Institutional Review Board approval from the Department of Psychology at Northwest Normal University. We received the written consent of all participants before testing began. All procedures performed in studies involving human participants were by the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed consent Informed consent was obtained from all individual participants included in the study.

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